

Corrective Action Report — CAR-2025-001

Field	Value
Document ID	CAR-2025-001
Issue Date	2025-07-14
Author	M. Castaneda, Senior Reliability Auditor; reviewed by D. Pham, Compliance Lead
CAR Status	Open — Target Date Missed (3rd revision)
Subject	Q1 2025 Budget Freeze Deferred 14 Preventive Maintenance Work Orders at NW-PowerPool; Direct Upstream Cause of Spring Vibration Cascade
Affected Region / Site / Asset	NW-PowerPool / NW-PowerPool-WIND-014, NW-PowerPool-WIND-024 / WIN-10058, WIN-10016, WIN-10011, WIN-10131, SUB-10641
Severity	Critical
Owner	T. Harlan, Director of Reliability — NW-PowerPool
Original Target Close Date	2025-04-30 (missed)
Revised Target Close Date	2025-09-30 (missed); current revision 2026-01-31
Distribution	VP Operations; NW-PowerPool Plant Manager; Procurement; Internal Audit; Engineering Standards Committee

1. Finding

A formal review of Computerized Maintenance Management System (CMMS) work-order ledgers covering the period 2025-01-06 through 2025-03-31 confirms that fourteen (14) preventive maintenance (PM) work orders scheduled at NW-PowerPool-WIND-014 and NW-PowerPool-WIND-024 were administratively deferred under the Q1 2025 capital-and-operating expense freeze (Finance Directive FD-2025-002, dated 2024-12-19). Of the 14 deferred PMs, 11 were rescheduled past their Cascadia Standard

MS-2025-03 maintenance-window tolerance (± 30 days). Three were closed out as "completed by rescheduling" without on-site service being performed.

The deferred work scope included bearing vibration spectral analysis, gearbox oil-spectrometer sampling, and substation breaker timing tests — the precise failure modes observed during the May–June 2025 Spring Vibration Cascade.

2. Evidence

2.1 Deferred work orders (CMMS records)

Work Order	Originally Scheduled	Asset	Scope	Disposition
WO-2025-00417	2025-01-22	WIN-10058	Gearbox vibration spectral scan + oil sample	Deferred → never rescheduled
WO-2025-00418	2025-01-22	WIN-10058	Main-bearing endoscopic inspection	Deferred → rescheduled 2025-06-04 (post-failure)
WO-2025-00431	2025-01-29	WIN-10016	PM-2 bearing service + grease replacement	Deferred → closed administratively
WO-2025-00466	2025-02-05	WIN-10016	Pitch system vibration test	Deferred → rescheduled 2025-07-11
WO-2025-00472	2025-02-12	WIN-10011	Gearbox oil-spectrometer (CSM-7 sample)	Deferred → never rescheduled
WO-2025-00473	2025-02-12	WIN-10131	Gearbox oil-spectrometer (CSM-7 sample)	Deferred → never rescheduled
WO-2025-00488	2025-02-19	SUB-10641	Breaker timing + dielectric test	Deferred → rescheduled 2025-08-22

Work Order	Originally Scheduled	Asset	Scope	Disposition
WO-2025-00489	2025-02-19	SUB-10641	Transformer DGA sample	Deferred → closed administratively
WO-2025-00503	2025-02-26	WIN-10058	Yaw-system torque audit	Deferred → never rescheduled
WO-2025-00517	2025-03-05	WIN-10016	Generator coupling alignment	Deferred → rescheduled 2025-07-25
WO-2025-00531	2025-03-12	WIN-10131	Blade-root bolt torque audit	Deferred → never rescheduled
WO-2025-00532	2025-03-12	WIN-10011	Blade-root bolt torque audit	Deferred → never rescheduled
WO-2025-00549	2025-03-19	SUB-10641	Insulation resistance + Doble test	Deferred → rescheduled 2025-09-09
WO-2025-00558	2025-03-26	WIN-10058	Main-shaft alignment	Deferred → rescheduled 2025-06-04 (post-failure)

2.2 Linked failure events (CMMS outage ledger)

- OUT-2025-0433 (2025-05-02, WIN-10058, soc_drift, 227 h, 2,506.8 MWh lost) — preceded by deferred WO-2025-00417 and WO-2025-00503.
- OUT-2025-0345 (2025-04-27, WIN-10016, soc_drift, 219 h, 2,418.4 MWh lost) — preceded by deferred WO-2025-00431 and WO-2025-00466.
- OUT-2025-0336 (2025-04-01, WIN-10016, overheating, 175 h, 1,932.5 MWh lost) — preceded by deferred WO-2025-00431.
- OUT-2025-1405 (2025-05-26, TRA-10561, vibration_excess, 158 h, 2,615.1 MWh lost) — adjacent maintenance unit; deferred bearing PM cited in RCA-2025-002.
- OUT-2025-1627 (2025-06-28, SUB-10641, efficiency_decline, 176 h, 1,943.6 MWh lost) — preceded by deferred WO-2025-00488 and WO-2025-00489.

2.3 Financial exposure

Period A (May–June 2025) emergency vendor invoices totaled **\$4.6M** (Apex Emergency Grid Solutions, under EPA-2024-11 activation; cross-reference EMR-2025-001 and EMR-2025-002). Avoidance value of the 14 deferred PMs, computed at CPS scheduled-work rates, was approximately **\$612K**. Net financial

penalty of the deferral exceeds **\$4.0M** before counting lost-generation revenue (27,400 MWh region-wide in Period A; see OIR-2025-003).

3. Root Cause Hypothesis

- **Primary:** Finance Directive FD-2025-002 imposed an undifferentiated freeze across capital and operating budget categories. PM execution at NW-PowerPool was funded through the operating budget; no exception process existed for safety- or reliability-critical PMs.
- **Contributing:** The CMMS deferral workflow allowed deferral approval at the site-supervisor level, with no escalation gate to Engineering for assets on the "watch list" (WIN-10016 had been added to the watch list in 2024 — see CAR-2024-118).
- **Contributing:** No reliability-risk score was attached to deferred PMs at the time of approval. The system did not flag that 11 of the 14 deferrals were on assets with three or more forced outages in the trailing twelve months.

4. Conflict With Existing Guidance

- The deferrals violate **Cascadia Standard MS-2025-03 §4.3**, which states PM intervals on Tier-2 (watch-list) assets shall not exceed +30 days from scheduled date without Engineering concurrence.
- The deferrals are consistent in form (but not substance) with **Finance Directive FD-2025-002**, which contained no carve-out for reliability-critical work.
- Action **CAR-2024-118** (issued December 2024, "Preventive Maintenance Cadence Inadequate For Aging Wind Assets") had already flagged WIN-10016 and SUB-10641 as inadequately serviced. CAR-2024-118 remains open.

5. Recommended Corrective Action

1. **Issue Standing Exception SE-2025-014** to Finance Directive FD-2025-002 covering all Tier-1 and Tier-2 reliability-critical PMs. Exception authority to rest with the Director of Reliability, not site supervisors.
2. Modify CMMS deferral workflow to require Engineering co-sign for any PM deferral on an asset with two or more forced outages in trailing 12 months. Hard system gate, not advisory.
3. Execute the eight (8) deferred WOs that were "never rescheduled" within 60 days of CAR re-issue. List in §6 below.
4. Reconcile WO-2025-00472 and WO-2025-00473 (oil-spectrometer samples) against the ambiguity flagged in CAR-2025-003 — sample frequency for CSM-7 differs between CPS and Apex execution.

5. Add a recurring quarterly Reliability & Compliance review of "administratively closed" PMs (the three orders closed without on-site work being performed).

6. Action Owner & Timeline

Sub-action	Owner	Target Date
Standing Exception SE-2025-014 issued	T. Harlan / CFO Office	2025-08-31 — missed
CMMS workflow change (Engineering co-sign gate)	R. Vasquez (CMMS Admin)	2025-10-31 — missed
Execute 8 never-rescheduled WOs	NW-PowerPool Maintenance Supervisor	2025-12-15 — missed
Quarterly review process stood up	Reliability & Compliance team	2026-01-31 — open
Final closure (independent audit verification)	Internal Audit	2026-03-31

7. Implementation Status

Open — Status Yellow. As of 2026-05-01, 5 of 8 never-rescheduled work orders remain unexecuted (WO-2025-00472, -00473, -00503, -00531, -00532). The CMMS workflow modification has been scoped but not deployed pending vendor change request. Standing Exception SE-2025-014 was drafted in September 2025 but stalled in CFO Office review. This is the third revision of the target close date.

Cross-references: CAR-2024-118 (predecessor PM-cadence finding); CAR-2025-003 (CSM-7 ambiguity affecting WO-2025-00472/-00473); CAR-2025-006 (fatigue management during Period A); CAR-2025-009 (escalation policy non-compliance for WIN-10016, SUB-10641); OIR-2025-003 (Period A outage investigation); EMR-2025-001 and EMR-2025-002 (Period A Apex emergency invoices); RCA-2025-002 (TRA-10561 root cause).

End of report.